RadixMaths

A Level Mathematics

Practice Paper 3

Pure Mathematics

Level: **Challenge**

Much harder than a real exam, with fewer steps given to you. It uses only A Level content.
Few students will finish in 2 hours, and that is expected.

Time: 2 hours **Total marks: 100**

For students of every exam board. This paper covers only the pure mathematics that the Department for Education requires in every A Level Mathematics course, so it suits AQA (7357), Edexcel (9MA0), OCR A (H240) and OCR B MEI (H640). Questions may use content from Year 12 or Year 13.

You will need: a calculator, your exam board's formulae booklet, and paper to write on.

Instructions

- Answer **all** the questions, and label your answers to each part clearly.
- Show enough working to make your method clear. Answers without working may not gain full credit.
- Where a question says *show detailed reasoning*, or says that solutions relying on calculator technology are not acceptable, you must show every step of your working.
- Give inexact answers to three significant figures unless the question says otherwise. Use radians unless a question uses degrees.
- Diagrams are not drawn to scale.

Information

- The total mark for this paper is 100. The marks for each part are shown in brackets, for example **(3)**. Use them as a guide to how long to spend.
- The mark scheme and full worked solutions are in a separate document. Try the whole paper under timed conditions before you look at them.

1. Prove by contradiction that $\log_2 3$ is irrational. **(4)**

(Total for Question 1 is 4 marks)

2. Solve the simultaneous equations

$$\log_x y + \log_y x = \tfrac{5}{2}$$
$$xy = 27$$

where x and y are positive and not equal to 1. **(6)**

(Total for Question 2 is 6 marks)

3. The first three terms of a geometric sequence are, in order, the 1st, 4th and 13th terms of an arithmetic sequence whose common difference is not zero.

(a) Find the common ratio of the geometric sequence. **(3)**

The sum of the first four terms of the geometric sequence is 120.

(b) Find the sum of the first 20 terms of the arithmetic sequence. **(3)**

(Total for Question 3 is 6 marks)

4. A sequence is defined by

$$u_{n+1} = \frac{u_n - 1}{u_n + 1}, \qquad u_1 = 2$$

(a) Find u_2, u_3, u_4 and u_5, and describe the behaviour of the sequence. **(2)**

(b) Find the exact value of $\displaystyle\sum_{r=1}^{100} u_r$. **(2)**

(c) Prove that there is no real value of u_1 for which this sequence is constant. **(3)**

(Total for Question 4 is 7 marks)

5. Find the exact value of

$$\int_0^{\pi/2} \mathrm{e}^{2x} \cos x \, \mathrm{d}x$$

(6)

(Total for Question 5 is 6 marks)

6. Find the exact total area of the regions enclosed between the curves $y = \sin x$ and $y = \sin 2x$ for $0 \leqslant x \leqslant \pi$. **(7)**

(Total for Question 6 is 7 marks)

7. A circle touches both coordinate axes and passes through the point $P(9, 2)$.

(a) Find the equations of the two possible circles. **(5)**

(b) The two circles meet at P and at a second point Q. Find the exact length of PQ. **(3)**

(Total for Question 7 is 8 marks)

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8. The point $P\left(2p, \frac{2}{p}\right)$, where $p > 0$, lies on the curve with equation $xy = 4$. The tangent to the curve at P meets the x-axis at A and the y-axis at B.

Prove that the area of the triangle OAB is the same for every value of p, and that P is the midpoint of AB. **(6)**

(Total for Question 8 is 6 marks)

9. A median of a triangle is a line from a vertex to the midpoint of the opposite side.

Prove, using vectors, that the three medians of any triangle meet at a single point, and that this point divides each median in the ratio 2 : 1. **(6)**

(Total for Question 9 is 6 marks)

10. (a) Prove that

$$\cos 3\theta \equiv 4\cos^3\theta - 3\cos\theta$$

(3)

(b) Hence find the three solutions of the equation

$$8x^3 - 6x - 1 = 0$$

giving each in the form $\cos(k\pi)$, where k is a rational number. **(4)**

(c) Hence find the exact value of

$$\cos\frac{\pi}{9}\cos\frac{5\pi}{9}\cos\frac{7\pi}{9}$$

(2)

(Total for Question 10 is 9 marks)

11. The curve C has equation

$$x^2 + xy + y^2 = 12$$

(a) Find the coordinates of the points on C at which the tangent is parallel to the x-axis. **(5)**

(b) Find the exact greatest distance from the origin to a point on C. **(5)**

(Total for Question 11 is 10 marks)

12. (a) Show that the curve with equation

$$y = \frac{\ln x}{x}, \qquad x > 0$$

has exactly one stationary point. Find its exact coordinates and determine its nature. **(4)**

(b) Hence prove that $e^{\pi} > \pi^{e}$. **(3)**

(Total for Question 12 is 7 marks)

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13. By considering the binomial expansion of $(1-x)^{-\frac{1}{2}}$, find the exact value of

$$S = 1 + \frac{1}{4} + \frac{1\times 3}{4\times 8} + \frac{1\times 3\times 5}{4\times 8\times 12} + \frac{1\times 3\times 5\times 7}{4\times 8\times 12\times 16} + \cdots$$

(8)

(Total for Question 13 is 8 marks)

14. Water flows into a large tank, which is empty at time $t = 0$. The depth of the water, h metres, t minutes later is modelled by the differential equation

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{1-\sqrt{h}}{10}, \qquad 0 \leqslant h < 1$$

(a) Use the substitution $u = \sqrt{h}$ to show that

$$t = -20\left(\sqrt{h} + \ln\left(1-\sqrt{h}\right)\right)$$

(7)

(b) Find the time it takes for the depth of the water to reach 0.81 m. **(2)**

(c) Explain why, according to the model, the depth of the water never reaches 1 m. **(1)**

(Total for Question 14 is 10 marks)

TOTAL FOR PAPER IS 100 MARKS

END OF QUESTIONS

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